



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

November 2010
Assessment Task 1
Year 11

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- All answers to be given in simplified exact form unless otherwise stated.
- Marks may not be awarded for messy or badly arranged work

Total Marks – 80

- Attempt questions 1-5
- Start each new question in a separate answer booklet.
- Hand in your answers in 5 separate bundles:
 - Question 1,
 - Question 2,
 - Question 3,
 - Question 4 and
 - Question 5

Examiner: *A Ward*

Start a new booklet.

Question 1 (16 marks).	Marks
a) Determine whether the line $y = 3x + 7$ passes through the point $(27, 86)$.	1
b) What is the coefficient of $11x^4$?	1
c) Rewrite $\log_5 625 = x$ in exponential form (but do not solve for x).	1
d) State whether or not $\{(1, 2), (2, 4), (1, 5)\}$ is a function.	1
e) Without solving the equation $2x^2 - x - 3 = 0$, determine and justify whether the roots are: (i) Real or non-real. (ii) Equal or distinct. (iii) Rational or irrational.	3
f) Determine whether the following functions are odd, even or neither: (i) $f(x) = x^3 + 3x^2 + 7x$ (ii) $f(x) = \frac{2}{x^3}$ (iii) $f(x) = \frac{1}{x^5 + x^3}$	3
g) If the equation $x^2 + 3px + p$, where p is a non-zero constant, has equal roots; find the value of p .	2
h) The probabilities that Albert, Byron and Charlie will pass an examination are $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ respectively. What is the probability that Albert and Charlie will pass and Byron will fail?	2
i) Simplify first and find the value of x : $\log_{10} 5 + 2\log_{10} 8 - \frac{1}{2}\log_{10} 16 = \log_{10} x$	2

End of Question 1

Start a new booklet.

Question 2 (16 Marks).

Marks

- a)** A factory produces components of which 0.6% are defective. The components are packed in boxes of 10. A box is selected at random. Find the probability that the box contains exactly 2 defective components. 2
- b)** Differentiate the following with respect to x : 6
- (i) $x^5 - 4x^4 + 2x^2 - 7$
 - (ii) $(3x + 2)^7$
 - (iii) $x(2x^3 - 13)$
 - (iv) $\frac{8x^2 + 6}{x - 1}; x \neq 1$
- c)** Find the equation of the tangent to the curve $y = x^2 + 2x$, that is parallel to the line $y = 4x + 1$. 3
- d)** A parabola has vertex $(-2, -3)$ and directrix $y = -1$. Find the: 5
- (i) focal length
 - (ii) focus
 - (iii) axis of symmetry
 - (iv) equation of the parabola

End of Question 2

Start a new booklet.

Question 3 (16 marks).

Marks

a) Find the $\lim_{x \rightarrow 2} \frac{2x^2 - 8}{x - 2}$ 1

b) Find to 3 significant figures the value of x for which $5^x = 7$ 2

c) Sketch the regions satisfying the simultaneous inequations such that: 3

$$\{(x, y) : y \geq x^2\} \cap \{(x, y) : x + y < 3\}$$

d) Find A , B and C if $A(x-1)^2 + B(x-1) + C \equiv 3x^2 + 5x + 8$ 3

e) Differentiate the following from first principles: 3

$$f(x) = 3x^2 - 2x$$

f) Given that $y = x^2 + 5x - 3$ and $9x - y - 7 = 0$. 4

(i) Find any points of intersection.

(ii) Prove that $9x - y - 7 = 0$ is a tangent to $y = x^2 + 5x - 3$

End of Question 3

Start a new booklet.

Question 4 (16 marks).	Marks
a) Solve: $9 \times 3^{x-1} = \frac{1}{27}$	2
b) Find the exact solution to the equations: (i) $\ln(3x-7) = 5$ (ii) $3^x e^{7x+2} = 15$	4
c) The functions f and g are defined by: $f(x) = e^{2x} + 3$ $g(x) = \ln(x-1)$ Find $f[g(x)]$ and state its range.	3
d) Bag A contains 2 black and 5 white balls. Bag B contains 4 black and 6 white balls. One bag is selected at random and 2 balls taken from it, without replacement. What is the probability that one ball is black and one ball is white?	3
e) (i) Show that the locus of the point P, which moves so that its distance from A(1,2) is always three times its distance from B(5,6), is a circle. (ii) State its centre and radius.	4

End of Question 4

Start a new booklet.

Question 5 (16 marks).

Marks

a) Sketch the derivative of $y = |x|$. 2

b) The curve C has the equation: 9

$$y = x^2(x-6) + \frac{4}{x}; x > 0.$$

The points P and Q lie on C and have x co-ordinates 1 and 2 respectively.

(i) Show that the length of PQ is $\sqrt{170}$.

(ii) Show that the tangents to C at P and Q represent the same line.

(iii) Find an equation of the normal to C at P giving your answer in general form.

c) If a car dealership sets the price of their cars at \$28000, they will sell 54 cars. Every time they drop the price \$1000, 2 more cars will be sold. What should the price of their cars be set at to maximise income? 5

End of Question 5.

End of Examination.

Solutions Nov 2011 YR11 Assess Task 1.

16

(1) (a) $86 = 3 \times 27 + 7$ NO! (1)
 $86 = 88$

(b) 11 (1)

(c) $5^x = 625$ (1)

(d) $a=2, b=-1, c=-3$

$$\Delta = (-1)^2 - 4 \times 2 \times -3$$

$$= 1 + 24$$

$$= 25$$

(i) real (1)

(ii) distinct (1)

(iii) rational (1)

(f) (i) $f(-x) = (-x)^3 + 3(-x)^2 + 7(-x)$
 $= -x^3 + 3x^2 - 7x$
 $= -(x^3 - 3x^2 + 7x)$

$\neq f(x)$, no odd or even.
 or $\neq -f(x)$ neither (1)

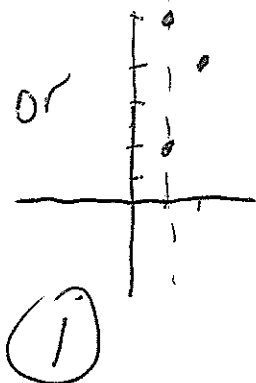
(ii) $f(-x) = \frac{2}{(-x)^3} = \frac{2}{-x^3} = -\frac{2}{x^3}$

So not even $\neq f(x)$

but $f(-x) = -f(x)$ odd (1)

(iii) $f(x) = \frac{1}{(-x)^5 + (-x)^3} = \frac{1}{-x^5 - x^3} = \frac{-1}{x^5 + x^3} \neq f(x)$ not even
 $= -f(x)$ odd (1)

(d) $1 < \frac{2}{5}$
 $2 < 4$
NO



(g) $x^2 + 3px + p$
has equal roots.

$$a = 1$$

$$b = 3p$$

$$c = p$$

$$\Delta = b^2 - 4ac = 0$$

$$9p^2 - 4 \times 1 \times p = 0$$

$$9p^2 - 4p = 0$$

$$p(9p - 4) = 0$$

$$p = 0 \text{ or } p = \frac{4}{9}$$

but $p \neq 0$ data so $p = \frac{4}{9}$ (2)

(h) $P(\text{Albert pass, Charlie pass, Byron fail})$

$$= \frac{1}{2} \times \frac{1}{\cancel{2}} \times \frac{\cancel{2}}{4}$$

$$= \frac{1}{8} \quad (2)$$

LHS (i) $\log 5 + 2\log 8 - \frac{1}{2}\log 16 = \log x$

$$\log 5 + \log 8^2 - \log 16^{\frac{1}{2}}$$

$$\log 5 + \log 64 - \log 4$$

$$\log \left(\frac{5 \times 64}{4} \right)$$

$$\log 80 = \log x$$

$$x = 80$$

(2)

2 UNIT SOLUTIONS

QUESTION 2

a) Defective 0.006

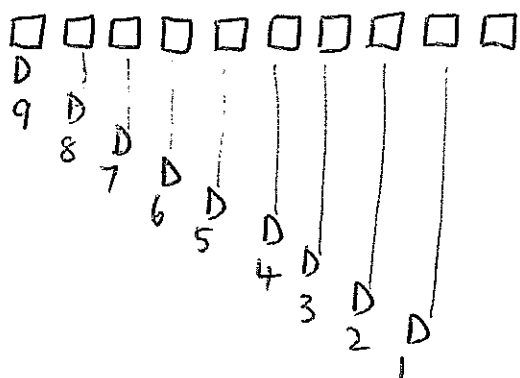
Non-Defective 0.994

□ □ □ □ □ □ □ □ □ □

2 Defective means 8 Non-Defective.

$$\text{i.e. } 0.006 \times 0.006 \times (0.994)^8$$

Also, how many different positions can defective ones be arranged, amongst the ten in the box



$$\text{i.e. } 9+8+7+6+5+4+3+2+1=45$$

$$\therefore P(E) = 45 \times 0.006 \times 0.006 \times (0.994)^8$$

$$[OR {}^{10}C_2 \times 0.006^2 \times 0.994^8]$$

b) (i) $5x^4 - 16x^3 + 4x$

(ii) $21(3x+2)^6$

(iii) $2x^3 - 13 + x \cdot 6x^2$
 $= 2x^3 - 13 + 6x^3$
 $= 8x^3 - 13$

(iv) $\frac{(x-1) \cdot 16x - (8x^2+6)}{(x-1)^2}$

$$= \frac{16x^2 - 16x - 8x^2 - 6}{(x-1)^2}$$

$$= \frac{8x^2 - 16x - 6}{(x-1)^2}$$

$$= 2 \frac{(4x^2 - 8x - 3)}{(x-1)^2}$$

c) $y = x^2 + 2x$

$$\frac{dy}{dx} = 2x + 2$$

but $m=4$ as lines parallel

$$\therefore 2x + 2 = 4$$

$$2x = 2$$

$$x = 1$$

and so $y = 3$.

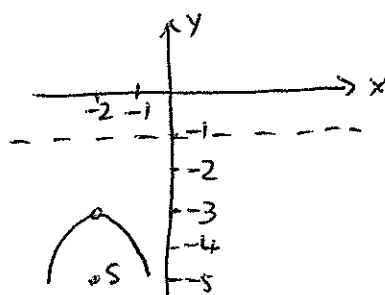
Eqn of tang: $m=4$ (1,3)

$$y - 3 = 4(x - 1)$$

$$y - 3 = 4x - 4$$

$$\boxed{y = 4x - 1}$$

d)



(i) Focal length = 2

(ii) Focus $(-2, -5)$

(iii) $x = -2$

$$(x+2)^2 = -4a(y+3)$$

$$(x+2)^2 = -8(y+3)$$

Marking Scale.

a) 0.006×0.006 1 mark.

0.06×0.06
OR the like $\frac{1}{2}$ mark

$0.006^2 \times 0.994^8$ $1\frac{1}{2}$ marks

$(0.994)^8$ 1 mark

b) (i) 1 mark (no half marks)

(ii) 1 mark (" " ")

$7(3x+2)^6$, 3 is OK.

(iii) One part of product
rule correct - 1 mark
(no half marks)

(iv) One error - 1 mark.
(no half marks)

c) $\left. \begin{array}{l} 2x+2=4 \\ \text{OR } y_1=2x+2 \end{array} \right\}$ 1 mark

Point (1,3) 1 mark

Eqn $y=4x-1$ 1 mark.
(No half marks)

d) (i) $a=-2$ $\frac{1}{2}$ mark.

(ii) no half marks.

(iii) -2 $\frac{1}{2}$ mark

$y=-2$ $\frac{1}{2}$ mark.

(iv) Leaving out the '-' sign
- (minus 1 mark)

Using focus instead of vertex
- (minus 1 mark)

1 2 3 3 3 4

Solution to Question (3)

$$(a) \lim_{x \rightarrow 2} \frac{2(x+2)(x/2)}{(x/2)}$$

$$[1] = \boxed{8} \quad \text{---} *$$

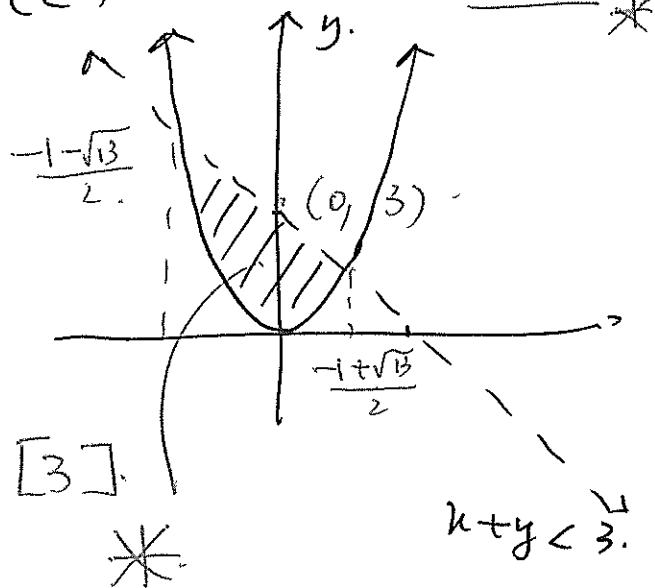
$$(b) \quad 5^x = 7$$

$$x \log_{10} 5 = \log_{10} 7$$

$$\therefore x = \frac{\log_{10} 7}{\log_{10} 5}$$

$$[2] \quad \boxed{x = 1.21} \quad \text{---} *$$

$$(c) \quad \text{---} *$$



$$(d) \quad A(x-1)^2 + B(x-1) + C$$

$$\equiv 3x^2 + 5x + 8$$

$$x=1, \quad \boxed{C=16}$$

$$\text{coeff of } x^2 \quad \boxed{A=3}$$

$$x=0$$

$$8 = A - B + C \quad [3]$$

$$8 = 3 - B + 16$$

$$\therefore -B = -11$$

$$\therefore \boxed{B=11}$$

$$(e) \quad f(x+h)$$

$$= 3(x+h)^2 - 2(x+h)$$

$$= 3[x^2 + 2xh + h^2]$$

$$- 2x - 2h$$

$$= 3x^2 + 6xh + 3h^2$$

$$- 2x - 2h$$

$$\therefore \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \frac{6x\cancel{h} + 3h^2 - 2h}{\cancel{h}}$$

$$\lim_{h \rightarrow 0} = 6x - 2. \quad [3]$$

$$(f). \quad 9x - 7 = x^2 + 5x - 3$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$\therefore x = 2, \quad y = 11$$

(i) $(2, 11)$ is pt of intersection.

$$x^2 - 4x + 4 = 0$$

$$(ii) \quad \Delta = 0 \quad [4]$$

$$\therefore y = 9x - 7$$

is a tangent to

$$y = x^2 + 5x - 3$$

Since $\nexists$ only a solution

QUESTION 4

$$(a) \quad 3^2 \times 3^{x+1} = 3^{-3}$$

$$x+1 = -3$$

$$x = -4$$

$$(b) \quad (i) \quad 3x - 7 = e^5$$

$$x = \frac{1}{3}(e^5 + 7)$$

$$(ii) \quad \ln(3^x e^{7x+2}) = \ln 15$$

$$\ln 3^x + \ln e^{7x+2} = \ln 15$$

$$x \log_e 3 + 7x + 2 = \log_e 15$$

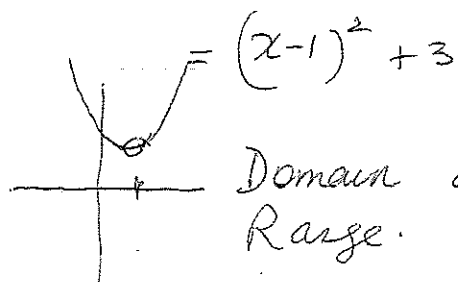
$$x(\log_e 3 + 7) = \ln 15 - 2$$

$$x = \frac{\ln 15 - 2}{\ln 3 + 7}$$

$$(c) \quad g(x) = \ln(x-1)$$

$$f[g(x)] = e^{2 \ln(x-1)} + 3$$

$$= e^{\log_e(x-1)^2} + 3$$



Domain all real x .
Range: $y > 3$.

$$(A) \quad 2B \quad SW \quad (B) \quad 4B \quad 6W \quad B$$

$$\begin{array}{c} \frac{2}{7} B \\ \frac{5}{7} W \end{array} \quad \begin{array}{c} \frac{1}{6} B \\ \frac{5}{6} W \end{array} \quad \begin{array}{c} \frac{4}{10} B \\ \frac{6}{10} W \end{array}$$

$$\frac{1}{2} \left(\frac{2}{7} \times \frac{5}{6} + \frac{5}{7} \times \frac{1}{6} \right) \quad \frac{1}{2} \left(\frac{4}{10} \times \frac{6}{9} + \frac{6}{10} \times \frac{4}{9} \right)$$

$$\frac{5}{21} + \frac{4}{15} = \frac{53}{105}$$

$$\frac{1}{2} \times \frac{{}^2C_1 \times {}^5C_1}{{}^7C_2} + \frac{1}{2} \times \frac{{}^4C_1 \times {}^6C_1}{{}^{10}C_2}$$

$$= \frac{53}{105}$$

$$(e) \quad \sqrt{(x-1)^2 + (y-2)^2} = 3\sqrt{(x-5)^2 + (y-6)^2}$$

$$x^2 - 2x + 1 + y^2 - 4y + 4 = 9(x^2 - 10x + 25 + y^2 - 12y + 36)$$

$$8x^2 - 88x + 8y^2 - 104y + 544 = 0$$

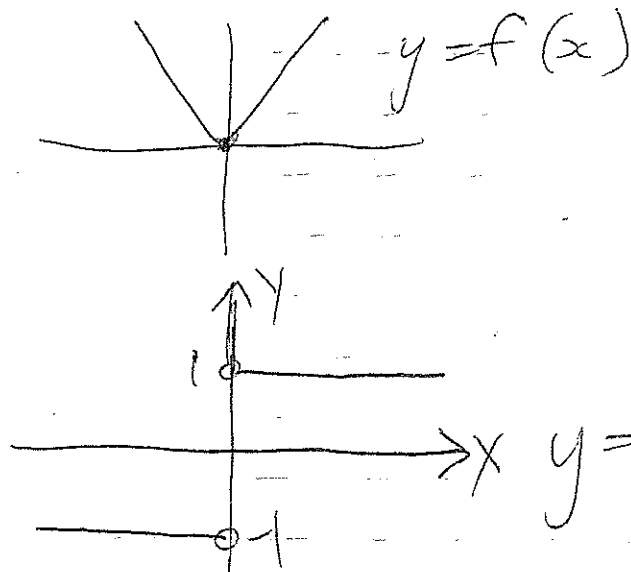
$$x^2 - 11x + \left(\frac{11}{2}\right)^2 + y^2 - 13y + \left(\frac{13}{2}\right)^2 = -68 + 12\frac{1}{2}$$

$$\left(x - \frac{11}{2}\right)^2 + \left(y - \frac{13}{2}\right)^2 = \frac{9}{2}$$

$$\text{Circle } C \left(\frac{11}{2}, \frac{13}{2}\right) \quad r = \frac{3\sqrt{2}}{2}$$

5. Q5. 11 Maths

(a) $y = |x|$



(b) $y = x^2(x-6) + \frac{4}{x}$; $x > 0$

P, Q lie on curve

$P(1, y_P)$

$Q(2, y_Q)$

(i) Show $PQ = \sqrt{170}$.

When $x = 1$, $y = 1(-5) + \frac{4}{1} = -1$
 $\Rightarrow P(1, -1)$ ✓

When $x = 2$, $y = 4(-4) + \frac{4}{2} = -14$
 $\Rightarrow Q(2, -14)$ ✓

$d_{PQ} = \sqrt{(2-1)^2 + (-14+1)^2}$
 $= \sqrt{1 + 169} = \sqrt{170}$ ✓

(ii) $y' = x^2 \cdot 1 + (x-6)2x - \frac{4}{x^2}$
 $= 3x^2 - 12x - \frac{4}{x^2}$
 Point P $y'(1) = 1 + -10 - 4 = -13$ ✓
 Point Q $y'(2) = 4 - 16 - 1 = -13$ ✓

(ii) (cont) Then, ^{eqn of} tangent at $P(1, -1)$ $m = -13$

is $y + 1 = -13(x - 1)$

$$y + 1 = -13x + 13$$

$$y = -13x + 12 \quad (1) \quad \checkmark$$

Eqn of tangent at $Q(2, -14)$ and $m = -13$

~~y + 14~~ $y + 14 = -13(x - 2)$
 $= -13x + 26$

$$y = -13x + 12 \quad (2) \quad \checkmark$$

$m_1 = m_2 \therefore$ parallel + same line

(iii) Normal at P. $\Rightarrow m = \frac{1}{13} \quad \checkmark$

$$y + 1 = \frac{1}{13}(x - 1)$$

$$13y + 13 = x - 1$$

$$-x + 13y + 14 = 0 \quad \# \quad \checkmark$$

(c)

Price	n (no sold)
28000	54
28000 - 1000	54 + 2
28000 - 2x1000	54 + 2x2

$x =$ no of \$1000 reductions

$$\text{Income} = \text{Price} \times n$$

$$I = (28000 - 1000x)(54 + 2x) \quad \checkmark$$

$$I = 1512000 + 56000x - 54000x - 2000x^2$$

$$I = 1512000 + 2000x - 2000x^2$$

$$I' = 2000 - 4000x \quad \checkmark$$

For t.p.'s $I' = 0 \Rightarrow -4000x = -2000$
 $x = \frac{1}{2}$

$$I'' = -4000 \Rightarrow \text{max at } x = \frac{1}{2}$$

or T.P. at $x = -\frac{b}{2a}$, (quadratic)

$$= \frac{-2000}{-4000}$$

$$x = \frac{1}{2} \text{ and } a < 0 \quad \checkmark \quad \checkmark \quad \therefore \text{max}$$

When $x = \frac{1}{2}$, $I = \$1513500$.

$\therefore$ Price should be set at

$$\begin{aligned} & \$28000 - 1000 \times \frac{1}{2} \\ & = \underline{\underline{\$27500}} \quad \checkmark \end{aligned}$$